SYNTHETIC GRAPH SOURCE {COUNT} $&

Musi ni sa athu u renga kana u thoma

Shumisani zwe na zwi vhona bulasini na mitevhe mipfufhi ya ṱhoḓisiso. Khwaṱhisedzani zwine na nanga henefho musi ni sa athu u renga.

Tshigwada tsha kilima tsha ṱhoḓisiso: Fhethu ha u lima a hu athu u ṱalutshedzea. Tshigwada itshi a si nyeletshedzo yo khwaṱhisedzwaho ya bulasi iḽi.

Zwe na vhiga nga bulasi iḽi

Maḓi nga khalaṅwaha yo oma

A zwo ngo rekhodwa. Fhedzisani ṱhoḓisiso ya fhethu ni khwaṱhisedze maḓi ni sa athu u nanga pulane ine ya ṱoḓa zwishumiswa zwinzhi.

Murotholo kha bulasi iḽi

Zwe zwa vhonwa nga murotholo wa henefho a zwo ngo rekhodwa. Zwikati zwa ṅwedzi zwa sathelaithi a zwi koni u khwaṱhisedza maḓuvha a murotholo wa u thoma na wa u fhedza.

Ḓuvha na u bva ha maḓi mavuni

Ḓuvha: a zwo ngo rekhodwa.

U bva ha maḓi: a zwo ngo rekhodwa. Mvelelo dza u ṱola mavu na ṱhoḓea dza tshiliṅwa tshiṅwe na tshiṅwe zwi kha ḓi ṱoḓa u ṱolwa.

Rekhodo dza ṅwaha wa u vhiga - hu si mbonelaphanḓa

Zwivhalo na miṅwedzi zwa ṅwaha wo fhiraho wa u vhiga zwi ṱalutshedza zwo vhigwaho. A zwi fulufhedzisi zwiḽiwa kha pulane i ḓaho.

Original eggs {year} $&

Tshivhalo tsha ṅwaha tsho vhigwaho nga mulimi tsha 2025: 240 eggs.

Zwo shumiswaho muṱani: 0 eggs. Zwo rengiswaho: a zwo ngo rekhodwa.

Miṅwedzi ya khaṇo yo vhigwaho: Fulw, Fwan. Izwi ndi zwe zwa vhonwa kha ṱhoḓisiso, hu si mbonelaphanḓa ntswa ya vhubveledzi.

Ṱolani yuniti na zwivhalo zwo rekhodwaho ni sa athu u shumisa mutaladzi uyu.

Fish

Tshivhalo tsha ṅwaha tsho vhigwaho nga mulimi tsha 2025: a zwo ngo rekhodwa.

Zwo shumiswaho muṱani: a zwo ngo rekhodwa. Zwo rengiswaho: a zwo ngo rekhodwa.

Miṅwedzi ya khaṇo yo vhigwaho: a zwo ngo rekhodwa. Izwi ndi zwe zwa vhonwa kha ṱhoḓisiso, hu si mbonelaphanḓa ntswa ya vhubveledzi.

Ṱolani yuniti na zwivhalo zwo rekhodwaho ni sa athu u shumisa mutaladzi uyu.

Miṅwe miroho

Tshivhalo tsha ṅwaha tsho vhigwaho nga mulimi tsha 2025: 12 kg.

Zwo shumiswaho muṱani: 7 kg. Zwo rengiswaho: 3 kg.

Miṅwedzi ya khaṇo yo vhigwaho: Phan, Nyen. Izwi ndi zwe zwa vhonwa kha ṱhoḓisiso, hu si mbonelaphanḓa ntswa ya vhubveledzi.

Ṱolani yuniti na zwivhalo zwo rekhodwaho ni sa athu u shumisa mutaladzi uyu.

A i athu ṱolwa nga muḓivhi wa zwa vhulimi — ndi tsumbanḓila ya u pulana, a si ndaela. I sumbedzeni muṱuṱuwedzi waṇu wa zwa vhulimi, ni ite ndingo ya mavu musi ni sa athu shumisa tshelede kha mavu.
Ṱhalutshedzelo ya mvetamveto: a i athu u sedzwa nga muthu a ambaho luambo zwavhuḓi. Sedzani tshiko tsha Luisimane phanḓa ha u tevhedza zwine zwa ambiwa.

SYNTHETIC GRAPH SOURCE {COUNT} $&

Zwine zwa nga nangwa zwa bulasi - zwi khou bvela phaṋda

Shumisani zwe na zwi vhona bulasini na mitevhe mipfufhi ya ṯhoḓisiso. Khwaṯhisedzani zwine na nanga henefho musi ni sa athu u renga.

Zwibveledzwa zwa zwifuwo na ṯhogomelo

Ṋotshi: Honeybee (managed hives)

Zwifhaṱo zwa 1 zwi re hone; zwa 2 zwo dzinginywaho. Bindu ḽo nangwaho: mutoli. A hu humbulelwi tshivhalo tsha zwifuwo kana mvelelo.

Nyeletshedzo ya u renga: yelanyisani lushaka/zwifuwo na tshibveledzwa, furu ya henefho, kilima na vhukoni ha ṯhogomelo; khwaṯhisedzani na mueletshedzi wa henefho wa zwifuwo kana u fuwa ṉotshi.

Khaṉo i ḓitika nga tshigwada tsha ṉotshi, maluvha na mvula ya henefho. Ṯolani u vhibva ni siele tshigwada zwiḽiwa; a hu na khalaṅwaha yo tiwaho ya lushaka ya mutoli.

Tshifhinga tsha tshibveledzwa tsha u thoma: a tshi na tshiko kha bindu iḽi.

Ndaedzi ya vhubveledzi: 25 kg ya mutoli/nga nnḓu ya ṉotshi (hive)/nga ṅwaha. From a 2008 national postal survey of 112 SABIO-member beekeepers (22.4% response rate) managing an estimated 19,520 hives at 2,094 apiary sites; figure is self-reported and un-weighted. Table 7 (same doc, p.13-14) shows wide variance by region/operation-focus/size for the same year: e.g. 'All' region, honey-focused, all-size operations = 25.1-26.9 kg/hive/yr across 3 survey sites; Western Cape honey-focused = 23.6-27.9 kg/hive/yr; Western Cape pollination-focused, small operations = only 7.0-7.5 kg/hive/yr. The same document also quotes an earlier, more rigorously validated study: 'Their average reported yield is very low, at just 13.4 kilograms per hive' for Western Cape hives (Allsopp & Cherry 2004, cited p.2 of this working paper), and historical national bee-census data of 8.3 kg/hive/annum (1911-1988 average) rising to 24 kg/hive in the 1974/5 census (same doc, p.1). No single number can be called definitive; 25 kg/hive/yr is used as the document's own headline national estimate.

Ṯolani ndaedzi dzo ṱanḓavhuwaho dza bindu dza zwiḽiwa na madzulo. Yelanyisani mukovhe wa zwiḽiwa na tshikhala na tshiimo tsha tshifuwo na sisṱeme ya madzulo; ṉeani maḓi o kunaho a fulufhedzeaho.

Veani miṱa ya ṉotshi kule na fhethu ha u tambela, ni ṉee maḓi o kunaho nahone ni shumise fhethu ho omaho. Humbelani mufuwi wa ṉotshi o ḓiṅwalisaho a sedzuluse u vhewa ho tsireledzeaho ha henefho phaṋḓa ha u engedza muṱa wa ṉotshi.

Sedzulusani tshigwada tshiṅwe na tshiṅwe tsha ṉotshi tshifhinga tshoṯhe u vhona tsumbo dza vhulwadze ha ṉotshi vhune ha tea u vhigiwa na u kavhiwa nga Cape-bee (Capensis).

Veani ni tsireledze miṱa ya ṉotshi kha vhuvhava na u tshinywa nga khole, zwine zwa vha iṅwe ya khaedu khulwane kha vhafuwi vha ṉotshi vha Afrika Tshipembe.

Register with the Department before beekeeping. Registration lasts 24 months from registration or renewal and must be renewed when it expires.

A beekeeper who fails to renew registration at expiry will be deregistered and face penalties under the Act.

Beekeeping activities and hired beekeeping services require a valid registration certificate. Check the certificate before using a beekeeper.

Every beehive must be clearly and legibly marked with the beekeeper's registration number.

Keep written records of colonies, apiary locations, diseases, disease-control measures and colony losses with their reasons.

Do not move honeybees across the Cape bee line, or keep Cape bees north of it, without written approval.

A i athu ṱolwa nga muḓivhi wa zwa vhulimi — ndi tsumbaṋḓila ya u pulana, a si ndaela. I sumbedzeni muṱuṱuwedzi waṉu wa zwa vhulimi, ni ite ndingo ya mavu musi ni sa athu shumisa tshelede kha mavu.
Ṯhalutshedzelo ya mvetamveto: a i athu u sedzwa nga muthu a ambaho luambo zwavhuḓi. Sedzani tshiko tsha Luisimane phaṋḓa ha u tevhedza zwine zwa ambiwa.

SYNTHETIC GRAPH SOURCE {COUNT} $&

Zwine zwa nga nangwa zwa bulasi - zwi khou bvela phanḓa

Shumisani zwe na zwi vhona bulasini na mitevhe mipfufhi ya ṱhoḓisiso. Khwaṱhisedzani zwine na nanga henefho musi ni sa athu u renga.

Zwibveledzwa zwa zwifuwo na ṱhogomelo (zwi khou bvela phanḓa)

Ṋotshi: Honeybee (managed hives) (zwi khou bvela phanḓa)

Report any suspected notifiable disease (American foulbrood is listed); officials can order an infected colony destroyed.

Tshiko: FAO: Beekeeping in Africa, Site selection (general African apiary guidance)

Tshiko: Department of Agriculture: Control Measures relating to honey-bees R.858 of 2013, consolidated with R.1511 of 22 November 2019, p. 6

Tshiko: Gums & Bees: A roadmap for landowners in South Africa (South African National Biodiversity Institute, SANBI), p. 21

Tshiko: Department of Agriculture: Control Measures relating to honey-bees R.858 of 2013, consolidated with R.1511 of 22 November 2019, control measures 2(1) and 2(4), p. 4

Tshiko: Department of Agriculture: Control Measures relating to honey-bees R.858 of 2013, consolidated with R.1511 of 22 November 2019, control measure 2(5), p. 4

Tshiko: Department of Agriculture: Control Measures relating to honey-bees R.858 of 2013, consolidated with R.1511 of 22 November 2019, control measures 2(8) and 2(9), p. 5

Tshiko: Department of Agriculture: Control Measures relating to honey-bees R.858 of 2013, consolidated with R.1511 of 22 November 2019, p. 5

Tshiko: Survey of Beekeeping in South Africa, CSSR Working Paper No. 221 (Conradie & Nortje, 2008, UCT Centre for Social Science Research), p. 14

Tshiko: FAO, Beekeeping in Africa, honey harvesting

A i athu ṱolwa nga muḓivhi wa zwa vhulimi — ndi tsumbanḓila ya u pulana, a si ndaela. I sumbedzeni muṱuṱuwedzi waṋu wa zwa vhulimi, ni ite ndingo ya mavu musi ni sa athu shumisa tshelede kha mavu.
Ṱhalutshedzelo ya mvetamveto: a i athu u sedzwa nga muthu a ambaho luambo zwavhuḓi. Sedzani tshiko tsha Luisimane phanḓa ha u tevhedza zwine zwa ambiwa.

SYNTHETIC GRAPH SOURCE {COUNT} $&

Zwine zwa nga nangwa zwa bulasi - zwi khou bvela phanḓa

Shumisani zwe na zwi vhona bulasini na mitevhe mipfufhi ya ṱhoḓisiso. Khwaṱhisedzani zwine na nanga henefho musi ni sa athu u renga.

Zwibveledzwa zwa zwifuwo na ṱhogomelo (zwi khou bvela phanḓa)

Danga: Nguni / crossbred beef cattle (communal and extensive veld)

Zwifhaṱo zwa 1 zwi re hone; zwa 2 zwo dzinginywaho. Bindu ḽo nangwaho: ṉama. A hu humbulelwi tshivhalo tsha zwifuwo kana mvelelo.

Nyeletshedzo ya u renga: yelanyisani lushaka/zwifuwo na tshibveledzwa, furu ya henefho, kilima na vhukoni ha ṱhogomelo; khwaṱhisedzani na mueletshedzi wa henefho wa zwifuwo kana u fuwa ṉotshi.

Tshifhinga tsha tshibveledzwa tsha u thoma: a tshi na tshiko kha bindu iḽi.

Ndaedzi ya vhubveledzi: 114.5–121.2 kg ya tshileme musi i tshi lumulwa/nga ṉamana. Table 5 gives weaning weight (WW) across four agro-ecological zones for the same nine Nguni communal herds: arid 114.51 kg (lowest), semi-arid 119.98 kg, dry sub-humid 117.91 kg, humid 121.21 kg (highest). The range recorded here spans this one study's own arid-to-humid zone means; it is not merged with any other paper's weaning-weight figures (other SA studies report different ranges for different systems - e.g. research-station or stud herds - and were deliberately excluded to avoid mixing systems). 826 Nguni calves from nine herds that the paper itself says were 'kept under extensive grazing system' in Limpopo Province - a genuine communal/extensive veld system matching this enterprise.

Ndaedzi ya vhutshilo ha vhubveledzi: miṅwaha ya 6.4. 2354 days / 365.25 = 6.44 years. 'Failure time' is the length of productive herd life for the 580 uncensored (fully-observed) records in the combined dataset; the dissertation itself defines productive herd life as 'the time interval between first calving date to culling date'. Combined data from three extensively-grazed Nguni stud herds, animals born 1968-2015.

Ṱolani ndaedzi dzo ṱanḓavhuwaho dza bindu dza zwiḽiwa na madzulo. Yelanyisani mukovhe wa zwiḽiwa na tshikhala na tshiimo tsha tshifuwo na sisṱeme ya madzulo; ṉeani maḓi o kunaho a fulufhedzeaho.

Zwitanga zwa kholomo a zwo ngo tea u ḓala u fhirisa: hu tea u vha na tshikhala tsho eḓanaho fhasi uri tshipuka tshiṅwe na tshiṅwe tshi kone u eḓela nga tshifhinga tshithihi.

Nga tshifhinga tsha u endedzwa, kholomo iṅwe na iṅwe i tea u vha na tshikhala tsha fhasi tshi si ho fhasi ha 1.4 m² uri u laḓa zwi songo tsikeledza nga maanḓa kana u sia tshikhala tshihulwane nga maanḓa.

Kholomo a dzo ngo tea u endedzwa tshifhinga tshi fhiraho awara dza 18 dzi sa athu u tsitswa dza ṉewa zwiḽiwa, maḓi na vhudzumbamo.

Ndi mulandu u tambudza kholomo nga nḓala, u dzi ṉea zwiḽiwa zwi sa eḓani, kana u dzi dzima maḓi kana zwiḽiwa.

Foot-and-mouth disease (FMD) is a controlled animal disease under the Animal Diseases Act: state Veterinary Services enforce isolation and movement restrictions.

Every cattle owner must register an official identification mark and brand or mark their cattle with it, under the Animal Identification Act.

Any suspected or confirmed brucellosis-related abortion in cattle must be reported to the State Veterinary office, as part of the national bovine brucellosis control scheme under the Animal Diseases Act.

No animal may be slaughtered anywhere but a registered abattoir, with the exception below.

A i athu ṱolwa nga muḓivhi wa zwa vhulimi — ndi tsumbanḓila ya u pulana, a si ndaela. I sumbedzeni muṱuṱuwedzi waṉu wa zwa vhulimi, ni ite ndingo ya mavu musi ni sa athu shumisa tshelede kha mavu.
Ṱhalutshedzelo ya mvetamveto: a i athu u sedzwa nga muthu a ambaho luambo zwavhuḓi. Sedzani tshiko tsha Luisimane phanḓa ha u tevhedza zwine zwa ambiwa.

SYNTHETIC GRAPH SOURCE {COUNT} $&

Zwine zwa nga nangwa zwa bulasi - zwi khou bvela phanḓa

Shumisani zwe na zwi vhona bulasini na mitevhe mipfufhi ya ṱhoḓisiso. Khwaṱhisedzani zwine na nanga henefho musi ni sa athu u renga.

Zwibveledzwa zwa zwifuwo na ṱhogomelo (zwi khou bvela phanḓa)

Danga: Nguni / crossbred beef cattle (communal and extensive veld) (zwi khou bvela phanḓa)

Slaughter for the farmer's own household consumption, or for cultural or religious purposes, is exempt from the abattoir rule.

Tshiko: Livestock Welfare Coordinating Committee, Code of Practice: Handling and Transport of Livestock (2018), s.1.2, p. 4

Tshiko: Livestock Welfare Coordinating Committee, Code of Practice: Handling and Transport of Livestock (2018), s.4.6.1, p. 7

Tshiko: Livestock Welfare Coordinating Committee, Code of Practice: Handling and Transport of Livestock (2018), s.7.1, p. 9

Tshiko: Animals Protection Act 71 of 1962, s.2(1)(c), p. 1

Tshiko: SAnews: KZN Foot and Mouth Disease boundaries extended (Dept of Agriculture statement)

Tshiko: Animal Identification Act 6 of 2002, s.7(1), p. 4

Tshiko: Marumo, Hlokwe & Kayoka-Kabongo, Seroprevalence of brucellosis in communal and smallholder cattle farming in North West Province, South Africa, Onderstepoort Journal of Veterinary Research (PMC10784229)

Tshiko: Meat Safety Act 40 of 2000, s.7(1)(a), p. 6

Tshiko: Meat Safety Act 40 of 2000, s.7(2)(a), p. 6

Tshiko: Mpofu, Ginindza, Siwendu, Nephawe & Mtileni (2016). Effect of agro-ecological zone, season of birth and sex on pre-weaning performance of Nguni calves in Limpopo Province, South Africa. Tropical Animal Health and Production 49:187-194 (PMC5203849), Table 5

Tshiko: Ngayo, M. (2017). Factors Affecting Productive Life and Fertility in Nguni Cows (MSc dissertation, University of the Free State), Table 5.1, p. 58

A i athu ṱolwa nga muḓivhi wa zwa vhulimi — ndi tsumbanḓila ya u pulana, a si ndaela. I sumbedzeni muṱuṱuwedzi waṋu wa zwa vhulimi, ni ite ndingo ya mavu musi ni sa athu shumisa tshelede kha mavu.
Ṱhalutshedzelo ya mvetamveto: a i athu u sedzwa nga muthu a ambaho luambo zwavhuḓi. Sedzani tshiko tsha Luisimane phanḓa ha u tevhedza zwine zwa ambiwa.

SYNTHETIC GRAPH SOURCE {COUNT} $&

Zwine zwa nga nangwa zwa bulasi - zwi khou bvela phanḓa

Shumisani zwe na zwi vhona bulasini na mitevhe mipfufhi ya ṱhoḓisiso. Khwaṱhisedzani zwine na nanga henefho musi ni sa athu u renga.

Zwibveledzwa zwa zwifuwo na ṱhogomelo (zwi khou bvela phanḓa)

Danga: Small-scale dairy cow (Jersey, Holstein or crossbred)

Zwifhaṱo zwa 1 zwi re hone; zwa 2 zwo dzinginywaho. Bindu ḽo nangwaho: mafhi. A hu humbulelwi tshivhalo tsha zwifuwo kana mvelelo.

Nyeletshedzo ya u renga: yelanyisani lushaka/zwifuwo na tshibveledzwa, furu ya henefho, kilima na vhukoni ha ṱhogomelo; khwaṱhisedzani na mueletshedzi wa henefho wa zwifuwo kana u fuwa ṋotshi.

Pulanani u ya nga u bebiwa ha vhukuma ha ṋamana/zwibudzana na u mamisa. Zwifuwo zwi sa mamisaho a zwi ṋei mafhi.

Ndaedzi ya tshibveledzwa tsha u thoma: vhege dza 121.8–130.5. Table 1, AFC column, rows in the table's order Holstein-Friesland Reg./Grade, Jersey Reg./Grade, Ayrshire Reg./Grade: 28, 30, 25, 28, 29, 32 months. The Grade rows used here are Holstein-Friesland 30 and Jersey 28 (Grade Ayrshire 32 is not used: the enterprise is Jersey or Holstein type). 28 x 4.35 = 121.8 weeks; 30 x 4.35 = 130.5 weeks. Milk begins at first calving, so age at first calving is the time to first product.

Ndaedzi ya vhubveledzi: 3840–5422 kg ya mafhi/nga kholomo ya tshisadzi/nga tshifhinga tsha u bveledza mafhi. Table 1 ('Comparison of number of cows, production and fertility in the main dairy breeds in South Africa'), Milk column, Grade rows: Jersey 3840 kg and Holstein-Friesland 5422 kg per lactation (Registered rows: 4468 and 6426 kg). The document says these are National Milk Recording Scheme herds, which 'have been shown to be better managed than those outside the scheme, with their performance per lactation being 50 % greater than that of the national herd', so a smallholder herd should expect less.

Ndaedzi ya vhutshilo ha vhubveledzi: miṅwaha ya 1.9–2. Holstein 739.33 and Jersey 696.81 days of productive life (first calving to culling) in South African milk-recorded herds, 2000-2019: 696.81/365.25 = 1.91 and 739.33/365.25 = 2.02 years.

Ndaedzi ya vhukale/sisṱeme: In hotter, humid KZN, KZN DARD describes Nguni × Jersey dairy trials at Makhathini to combine milk potential with local adaptation. Hand milking with the calf present highlights competition between household milk and calf growth. Discuss adapted stock and feeding locally; commercial milk-recording totals do not forecast this system. Record calving, lactation stage and milk available after calf needs.

Ṱolani ndaedzi dzo ṱanḓavhuwaho dza bindu dza zwiḽiwa na madzulo. Yelanyisani mukovhe wa zwiḽiwa na tshikhala na tshiimo tsha tshifuwo na sisṱeme ya madzulo; ṋeani maḓi o kunaho a fulufhedzeaho.

Zwitanga zwa kholomo a zwo ngo tea u ḓala u fhirisa: hu tea u vha na tshikhala tsho eḓanaho fhasi uri tshipuka tshiṅwe na tshiṅwe tshi kone u eḓela nga tshifhinga tshithihi.

Kholomo dza tsadzi a dzo ngo tea u laḓiwa uri dzi endedzwe arali maḓamu adzo o ḓala mafhi a vho mboḓa.

Kholomo a dzo ngo tea u endedzwa tshifhinga tshi fhiraho awara dza 18 dzi sa athu u tsitswa dza ṋewa zwiḽiwa, maḓi na vhudzumbamo.

Ndi mulandu u tambudza kholomo ya mafhi nga nḓala, u i ṋea zwiḽiwa zwi sa eḓani, kana u i dzima maḓi kana zwiḽiwa.

Kholomo dza tsadzi dzi re mafuloni kana gammbeni dzo vuleaho dzi ṱoḓa murunzi: zwifhaṱo zwa murunzi zwi tea u ṋea murunzi u si ho fhasi ha 4 u swika kha 5 m² nga kholomo.

Foot-and-mouth disease (FMD) is a controlled animal disease under the Animal Diseases Act: state Veterinary Services enforce isolation and movement restrictions.

Every cattle owner must register an official identification mark and brand or mark their cattle with it, under the Animal Identification Act.

A i athu ṱolwa nga muḓivhi wa zwa vhulimi — ndi tsumbanḓila ya u pulana, a si ndaela. I sumbedzeni muṱuṱuwedzi waṋu wa zwa vhulimi, ni ite ndingo ya mavu musi ni sa athu shumisa tshelede kha mavu.
Ṱhalutshedzelo ya mvetamveto: a i athu u sedzwa nga muthu a ambaho luambo zwavhuḓi. Sedzani tshiko tsha Luisimane phanḓa ha u tevhedza zwine zwa ambiwa.

SYNTHETIC GRAPH SOURCE {COUNT} $&

Zwine zwa nga nangwa zwa bulasi - zwi khou bvela phanḓa

Shumisani zwe na zwi vhona bulasini na mitevhe mipfufhi ya ṱhoḓisiso. Khwaṱhisedzani zwine na nanga henefho musi ni sa athu u renga.

Zwibveledzwa zwa zwifuwo na ṱhogomelo (zwi khou bvela phanḓa)

Danga: Small-scale dairy cow (Jersey, Holstein or crossbred) (zwi khou bvela phanḓa)

Dairy farmers are legally required to test their herds for bovine TB, mainly because of export requirements, so most dairy cattle in South Africa are tested.

Any suspected or confirmed brucellosis-related abortion in cattle must be reported to the State Veterinary office, under the national bovine brucellosis control scheme.

A cull cow may be slaughtered only at an abattoir, except for the farmer's own consumption or for cultural or religious purposes.

Tshiko: Livestock Welfare Coordinating Committee, Code of Practice: Handling and Transport of Livestock (2018), s.1.2, p. 4

Tshiko: Livestock Welfare Coordinating Committee, Code of Practice: Handling and Transport of Livestock (2018), s.6.13, p. 9

Tshiko: Livestock Welfare Coordinating Committee, Code of Practice: Handling and Transport of Livestock (2018), s.7.1, p. 9

Tshiko: Animals Protection Act 71 of 1962, s.2(1)(c), p. 1

Tshiko: Muller, C.J.C., The Dairy Farming Handbook (2017), Western Cape Department of Agriculture (Elsenburg), p. 129

Tshiko: SAnews: KZN Foot and Mouth Disease boundaries extended (Dept of Agriculture statement)

Tshiko: Animal Identification Act 6 of 2002, s.7(1), p. 4

Tshiko: Bovine tuberculosis / dairy herd testing study, PMC10369683

Tshiko: Marumo, Hlokwe & Kayoka-Kabongo, Seroprevalence of brucellosis in communal and smallholder cattle farming in North West Province, South Africa, Onderstepoort Journal of Veterinary Research (PMC10784229)

Tshiko: Meat Safety Act 40 of 2000, s.7(2)(a), p. 6

Tshiko: KZN DARD: Subsistence Dairying in KZN, Makhathini Research Station trials

Tshiko: Muller, C.J.C., The Dairy Farming Handbook (2017), Western Cape Department of Agriculture (Elsenburg), p. 10

Tshiko: Muller, C.J.C., The Dairy Farming Handbook (2017), Western Cape Department of Agriculture (Elsenburg), Table 21.1, p. 122

Tshiko: Gertenbach, W.D., Dairying in KwaZulu-Natal: Breeds of Dairy Cattle, Cedara Agricultural Development Institute, KZN Department of Agriculture and Rural Development, Table 1, p. 1

Tshiko: Cow Longevity and Reasons and Risk Factors for Culling in South African Holstein and Jersey Dairy Herds, Animals 2025 (PMC12560872)

A i athu ṱolwa nga muḓivhi wa zwa vhulimi — ndi tsumbanḓila ya u pulana, a si ndaela. I sumbedzeni muṱuṱuwedzi waṋu wa zwa vhulimi, ni ite ndingo ya mavu musi ni sa athu shumisa tshelede kha mavu.
Ṱhalutshedzelo ya mvetamveto: a i athu u sedzwa nga muthu a ambaho luambo zwavhuḓi. Sedzani tshiko tsha Luisimane phanḓa ha u tevhedza zwine zwa ambiwa.

SYNTHETIC GRAPH SOURCE {COUNT} $&

Zwine zwa nga nangwa zwa bulasi - zwi khou bvela phanḓa

Shumisani zwe na zwi vhona bulasini na mitevhe mipfufhi ya ṱhoḓisiso. Khwaṱhisedzani zwine na nanga henefho musi ni sa athu u renga.

Zwibveledzwa zwa zwifuwo na ṱhogomelo (zwi khou bvela phanḓa)

Khuhu: Broilers (commercial)

Zwifhaṱo zwa 1 zwi re hone; zwa 2 zwo dzinginywaho. Bindu ḽo nangwaho: ṉama. A hu humbulelwi tshivhalo tsha zwifuwo kana mvelelo.

Nyeletshedzo ya u renga: nangani lushaka lwa ṉama na pulane ya zwigwada ya muṉetshedzi; khwaṱhisedzani u dudedza zwikukwana, zwiḽiwa, maḓi na ndugiselelo dza u ṱhavha.

Pulanani zwiḽiwa zwo linganelaho, maḓi o kunaho, murunzi, muya na tsireledzo kha zwipuka zwi ḽaho zwiṅwe. Mphakelo wa makumba u ḓitika nga khuhu dzi kudzelaho, vhukale hadzo, mutakalo na ṱhogomelo; hoko i yoṱhe a i ṉei makumba.

Ndaedzi ya tshibveledzwa tsha u thoma: vhege dza 4.6–4.7. 32 days / 7 = 4.57 weeks; 33 days / 7 = 4.71 weeks. This enterprise has a single grow-out cycle ending in slaughter, so 'time to first product' = age at slaughter. South Africa, commercial industry average.

Ndaedzi ya vhubveledzi: 1.8–1.9 kg/nga tshiṉoni musi tshi tshi ṱhavhiwa (maḓuvha a 32–33). National commercial broiler industry average, South Africa, 2022. A separate breed-trial table (Table 4b, same document, p.39) gives named-strain 'Cumulative FI (kg)' and 'Body weight (kg)' at exactly 35 days: Arbor Acres Plus 2.29kg, Cobb 500 2.52kg, Indian River 2.30kg, Ross 308 2.30kg (cumulative feed intake 3.21-3.64kg) - a slightly older age (35 vs 32-33 days) and higher weight than the industry-wide average quoted above; not merged into the primary figure since it is a different (breed-standard trial) dataset, but used below for the feed-intake figure.

Ndaedzi ya vhukale/sisṱeme: Newer South African commercial reference: SAPA 2024 reports average slaughter at 30–31 days with bird weight 1.8–1.85 kg. The primary range above remains the earlier 2022 reference at 32–33 days. These are commercial bird weights, not dressed meat kg or a village flock target. Record hatch or purchase age, feeding system, actual harvest date and weighed product basis.

Ṱolani ndaedzi dzo ṱanḓavhuwaho dza bindu dza zwiḽiwa na madzulo. Yelanyisani mukovhe wa zwiḽiwa na tshikhala na tshiimo tsha tshifuwo na sisṱeme ya madzulo; ṉeani maḓi o kunaho a fulufhedzeaho.

Khuhu dza ṉama dzi dzulaho fhasi nḓuni: tshileme tsha khuhu tshi songo fhira 40 kg nga mitara ya tshikwea.

Zwikukwana zwi kha ḓi bva u thothonya zwi tea u ṉewa zwiḽiwa hu sa athu u fhira awara dza 24 zwi tshi bva u thothonya.

Zwikukwana zwi kha ḓi bva u thothonya zwi tea u kona u wana maḓi hu sa athu u fhira awara dza 24 zwi tshi bva u thothonya, nga u ṱavhanya u fhira zwenezwo musi hu tshi fhisa.

Maḓi o kunaho na zwiḽiwa zwo teaho zwi dzule zwi hone, uri khuhu dzi si vhe na nḓala na ḓora.

Nnḓu dza khuhu dza ṉama dzi ṱoḓa mpimo wa u dzhena na u tshimbila ha muya u si ho fhasi ha mitara dza khubiki dza 5 nga awara kha khilogireme iṅwe na iṅwe ya tshileme tsha khuhu dzi re nḓuni.

Kha u fuwa khuhu dza ṉama dzi tshimbilaho nnḓa, tshivhalo tshi dzulaho nḓuni a tsho ngo tea u fhira khuhu dza 15 nga mitara ya tshikwea ya vhupo ha fhasi.

An owner or manager whose animals are, or may be, infected with a controlled animal disease must report it to the state veterinary director immediately.

Newcastle disease and notifiable avian influenza are on the state list of controlled and notifiable animal diseases.

No animal may be slaughtered anywhere but an abattoir, with the exceptions below.

A i athu ṱolwa nga muḓivhi wa zwa vhulimi — ndi tsumbanḓila ya u pulana, a si ndaela. I sumbedzeni muṱuṱuwedzi waṉu wa zwa vhulimi, ni ite ndingo ya mavu musi ni sa athu shumisa tshelede kha mavu.
Ṱhalutshedzelo ya mvetamveto: a i athu u sedzwa nga muthu a ambaho luambo zwavhuḓi. Sedzani tshiko tsha Luisimane phanḓa ha u tevhedza zwine zwa ambiwa.

SYNTHETIC GRAPH SOURCE {COUNT} $&

Zwine zwa nga nangwa zwa bulasi - zwi khou bvela phanḓa

Shumisani zwe na zwi vhona bulasini na mitevhe mipfufhi ya ṱhoḓisiso. Khwaṱhisedzani zwine na nanga henefho musi ni sa athu u renga.

Zwibveledzwa zwa zwifuwo na ṱhogomelo (zwi khou bvela phanḓa)

Khuhu: Broilers (commercial) (zwi khou bvela phanḓa)

Slaughter for your own household, or for cultural or religious purposes, is exempt from the abattoir rule.

Tshiko: SAPA Code of Practice for Broiler Production (May 2012), 2.2 Housing of broilers in Floor Systems, p. 6

Tshiko: SAPA Code of Practice for Broiler Production (May 2012), p. 11

Tshiko: SAPA Code of Practice for Broiler Production (May 2012), p. 12

Tshiko: SAPA Code of Practice for Broiler Production (May 2012), p. 7

Tshiko: SAPA Code of Practice for Broiler Production (May 2012), p. 8

Tshiko: SAPA Code of Practice for Broiler Production (May 2012), 2.4 Free Range Broiler Production, p. 8

Tshiko: Animal Diseases Act 35 of 1984, s.11(1)(b)(ii), p. 9

Tshiko: Department of Agriculture, Forestry and Fisheries, Controlled and Notifiable Animal Diseases (factsheet), p. 1

Tshiko: Meat Safety Act 40 of 2000, s.7(1)(a), p. 6

Tshiko: Meat Safety Act 40 of 2000, s.7(2)(a), p. 6

Tshiko: SAPA Industry Profile 2024, section 7.9 Performance efficiency, p. 82

Tshiko: SAPA Industry Profile 2022, section 6.9 Performance efficiency, p. 79

Tshiko: ARC Climate-Smart Agriculture, module 11, printed pp. 468–480

A i athu ṱolwa nga muḓivhi wa zwa vhulimi — ndi tsumbanḓila ya u pulana, a si ndaela. I sumbedzeni muṱuṱuwedzi waṋu wa zwa vhulimi, ni ite ndingo ya mavu musi ni sa athu shumisa tshelede kha mavu.
Ṱhalutshedzelo ya mvetamveto: a i athu u sedzwa nga muthu a ambaho luambo zwavhuḓi. Sedzani tshiko tsha Luisimane phanḓa ha u tevhedza zwine zwa ambiwa.

SYNTHETIC GRAPH SOURCE {COUNT} $&

Zwine zwa nga nangwa zwa bulasi - zwi khou bvela phanḓa

Shumisani zwe na zwi vhona bulasini na mitevhe mipfufhi ya ṱhoḓisiso. Khwaṱhisedzani zwine na nanga henefho musi ni sa athu u renga.

Zwibveledzwa zwa zwifuwo na ṱhogomelo (zwi khou bvela phanḓa)

Khuhu: Village chickens (indigenous, free-range)

Zwifhaṱo zwa 1 zwi re hone; zwa 2 zwo dzinginywaho. Bindu ḽo nangwaho: makumba. A hu humbulelwi tshivhalo tsha zwifuwo kana mvelelo.

Buying advice: Potchefstroom Koekoek is a dual-purpose option; Venda, Ovambo and Naked Neck are other ARC-conserved options. No province-specific winning breed is claimed.

Pulanani zwiḽiwa zwo linganelaho, maḓi o kunaho, murunzi, muya na tsireledzo kha zwipuka zwi ḽaho zwiṅwe. Mphakelo wa makumba u ḓitika nga khuhu dzi kudzelaho, vhukale hadzo, mutakalo na ṱhogomelo; hoko i yoṯhe a i ṉei makumba.

Kuvhanganyani makumba tshifhinga tshoṯhe, vhulungani vhualo ha zwiṱaha ho kuna ho oma, ni fare/vhulunge makumba nga vhulondi u shumiswa muṱani kana u rengiswa.

Ndaedzi ya tshibveledzwa tsha u thoma: vhege dza 15.7–27.8. Free-range village chickens, KwaZulu-Natal; dry-environment birds started laying earlier (15.74 weeks) than wet-environment birds (27.78 weeks), difference flagged significant ('*') in the source table. Wide range reflects genuine dry/wet-season environmental variation reported by the source, not measurement uncertainty.

Ndaedzi ya vhubveledzi: 40.4–49.9 makumba/nga khuhu ya tshisadzi/nga ṅwaha. Free-range/scavenging village chickens, KwaZulu-Natal: dry (40.43) vs wet (49.91) environment, difference not significant ('NS'). A managed-system ARC Irene trial of named indigenous breeds (Grobbelaar, Sutherland & Molalakgotla 2010, Animal Genetic Resources 46: 25-32) reports far higher counts, now separately verified and retained in productionReferences; managed conditions cannot replace this village range.

Ndaedzi ya vhukale/sisṱeme: The ARC Irene trial compared Potchefstroom Koekoek, Venda, Ovambo and Naked Neck under commercial feed, water and controlled lighting. Mean eggs/hen in a 52-week cycle were 195.9, 153.7, 125.5 and 138.9 respectively; first-egg ages were 138.5, 139.0, 134.5 and 129.0 days from hatch, with no significant first-egg age difference among breeds. This is managed breed context, not the yield of scavenging hens in KZN. Match feeding and housing before using a benchmark.

Ṱolani ndaedzi dzo ṱanḓavhuwaho dza bindu dza zwiḽiwa na madzulo. Yelanyisani mukovhe wa zwiḽiwa na tshikhala na tshiimo tsha tshifuwo na sisṱeme ya madzulo; ṉeani maḓi o kunaho a fulufhedzeaho.

Kha maitele a mahayani a u fuwa fhethu ho vuleaho hu hulwane, khuhu dzi tea u ḓiwanela maḓi na zwiḽiwa.

Khuhu dza mahayani dzi ḓiṱoḓelaho zwiḽiwa dzi aluwa nga u ongolowa u fhira dzi fuwaho kha mabindu nahone dzi khomboni khulwane ya malwadze, zwilumi zwi dzulaho khadzo na zwipuka zwi dzi ḽaho.

Kha ṱhoḓisiso ya Limpopo na KwaZulu-Natal, 94.9% ya vhalimi vho vha vha tshi dzhenisa khuhu dzavho nḓuni vhusiku.

Zwe khuhu dza tsadzi dzi ḓitshimbilelaho dza vha dzi tshi ḓiwanela sa zwiḽiwa zwo vha zwi tshi eḓana u tshidza muvhili na u aluwa, fhedzi zwi sa eḓani u kudzela makumba: phurotheini kha khalaṅwaha dzoṯhe, calcium na phosphorus vhuria.

An owner or manager whose animals are, or may be, infected with a controlled animal disease must report it to the state veterinary director immediately.

Newcastle disease and notifiable avian influenza are on the state list of controlled and notifiable animal diseases.

No animal may be slaughtered anywhere but an abattoir, with the exceptions below.

A i athu ṱolwa nga muḓivhi wa zwa vhulimi — ndi tsumbanḓila ya u pulana, a si ndaela. I sumbedzeni muṱuṱuwedzi waṉu wa zwa vhulimi, ni ite ndingo ya mavu musi ni sa athu shumisa tshelede kha mavu.
Ṱhalutshedzelo ya mvetamveto: a i athu u sedzwa nga muthu a ambaho luambo zwavhuḓi. Sedzani tshiko tsha Luisimane phanḓa ha u tevhedza zwine zwa ambiwa.

SYNTHETIC GRAPH SOURCE {COUNT} $&

Zwine zwa nga nangwa zwa bulasi - zwi khou bvela phaṋḓa

Shumisani zwe na zwi vhona bulasini na mitevhe mipfufhi ya ṱhoḓisiso. Khwaṱhisedzani zwine na nanga henefho musi ni sa athu u renga.

Zwibveledzwa zwa zwifuwo na ṱhogomelo (zwi khou bvela phaṋḓa)

Khuhu: Village chickens (indigenous, free-range) (zwi khou bvela phaṋḓa)

Slaughter for your own household, or for cultural or religious purposes, is exempt from the abattoir rule.

Tshiko: ARC, Climate-Smart Agriculture Training Manual, Module 11: Poultry Production, p. 471

Tshiko: ARC, Climate-Smart Agriculture Training Manual, Module 11: Poultry Production, p. 468

Tshiko: Malatji, Tsotetsi, van Marle-Koster & Muchadeyi (2016), 'A description of village chicken production systems and prevalence of gastrointestinal parasites: Case studies in Limpopo and KwaZulu-Natal provinces of South Africa', Onderstepoort Journal of Veterinary Research 83(1)

Tshiko: Raphulu, Jansen van Rensburg & van Ryssen (2015), 'Nutrient content of crop contents of free ranging indigenous chickens raised under village conditions', South African Journal of Animal Science 45(2), p. 1

Tshiko: Animal Diseases Act 35 of 1984, s.11(1)(b)(ii), p. 9

Tshiko: Department of Agriculture, Forestry and Fisheries, Controlled and Notifiable Animal Diseases (factsheet), p. 1

Tshiko: Meat Safety Act 40 of 2000, s.7(1)(a), p. 6

Tshiko: Meat Safety Act 40 of 2000, s.7(2)(a), p. 6

Tshiko: Grobbelaar, Sutherland & Molalakgotla (2010), Egg production potentials of certain indigenous chicken breeds from South Africa: methods pp27–28, first-egg p29, Table 4 p31, p. 31

Tshiko: Tenza, Mhlongo & Chimonyo (2023), 'Village chicken production and egg quality in dry and wet, resource-limited environments in KwaZulu-Natal, South Africa', South African Journal of Animal Science 53(6), p. 5

Tshiko: ARC Climate-Smart Agriculture, module 11, printed pp. 468–480

Tshiko: FAO: Small-scale poultry production, egg marketing

A i athu ṱolwa nga muḓivhi wa zwa vhulimi — ndi tsumbaṋḓila ya u pulana, a si ndaela. I sumbedzeni muṱuṱuwedzi waṋu wa zwa vhulimi, ni ite ndingo ya mavu musi ni sa athu shumisa tshelede kha mavu.
Ṱhalutshedzelo ya mvetamveto: a i athu u sedzwa nga muthu a ambaho luambo zwavhuḓi. Sedzani tshiko tsha Luisimane phaṋḓa ha u tevhedza zwine zwa ambiwa.

SYNTHETIC GRAPH SOURCE {COUNT} $&

Zwine zwa nga nangwa zwa bulasi - zwi khou bvela phanḓa

Shumisani zwe na zwi vhona bulasini na mitevhe mipfufhi ya ṱhoḓisiso. Khwaṱhisedzani zwine na nanga henefho musi ni sa athu u renga.

Zwibveledzwa zwa zwifuwo na ṱhogomelo (zwi khou bvela phanḓa)

Khuhu: Laying hens (commercial)

Zwifhaṱo zwa 1 zwi re hone; zwa 2 zwo dzinginywaho. Bindu ḽo nangwaho: makumba. A hu humbulelwi tshivhalo tsha zwifuwo kana mvelelo.

Nyeletshedzo ya u renga: u bveledza makumba, humbelani khuhu ṱhukhu dzo haelwaho dzi re tsini na u thoma u kudzela kha muṋetshedzi wa khuhu dzi kudzelaho, na rekhodo dza vhukale, u fhiwa zwiḽiwa na u haelwa.

Pulanani zwiḽiwa zwo linganelaho, maḓi o kunaho, murunzi, muya na tsireledzo kha zwipuka zwi ḽaho zwiṅwe. Mphakelo wa makumba u ḓitika nga khuhu dzi kudzelaho, vhukale hadzo, mutakalo na ṱhogomelo; hoko i yoṯhe a i ṋei makumba.

Kuvhanganyani makumba tshifhinga tshoṯhe, vhulungani vhualo ha zwiṱaha ho kuna ho oma, ni fare/vhulunge makumba nga vhulondi u shumiswa muṱani kana u rengiswa.

Ndaedzi ya khalaṅwaha: With a constant 14–16 hour light day, commercial hens keep laying at a high rate all year. Ndaedzi iyi ya ndangulo ya vhubindudzi a si mphakelo wa makumba wa muṱa u linganaho nga mvelo.

Ndaedzi ya tshibveledzwa tsha u thoma: vhege dza 21. Vhukale ha khuhu u bva i tshi thothonywa; khwaṱhisedzani lushaka na u aluswa na muṋetshedzi. Khuhu dzi re tsini na u kudzela dzi thoma nga murahu kha mutevhe uyu.

Ndaedzi ya vhubveledzi: 407.1–428.6 makumba/nga khuhu ya tshisadzi yo vhewaho nḓuni (hen housed), mutevhe wa u kudzela u swika kha vhege dza 90. Table 4a's cumulative 'Eggs/hen housed' for five commercial strains; the same table gives 'Age at depop. weeks 90.0' for every strain. Commercial barn/cage average, not a free-range smallholder trial. Kept as the source gives it: a per-year rate would need a point-of-lay age from another document.

Ndaedzi ya vhukale/sisṱeme: SAPA 2024 commercial strain standards cumulative to 90 weeks: Dekalb Amberlink 429 eggs/hen housed, Dekalb Brown 428, Hy-Line Silver Brown 435, Hy-Line Brown 434 and Lohmann Brown Lite 413. These are full-cycle objectives, not eggs per calendar year or a smallholder forecast. Purchase age, actual onset of lay, surviving laying hens, feed, water and lighting need separate records.

Ṱolani ndaedzi dzo ṱanḓavhuwaho dza bindu dza zwiḽiwa na madzulo. Yelanyisani mukovhe wa zwiḽiwa na tshikhala na tshiimo tsha tshifuwo na sisṱeme ya madzulo; ṋeani maḓi o kunaho a fulufhedzeaho.

Kha maitele a u fuwa a tendelaho u tshimbila nnḓa, khuhu dza tsadzi dzi tea u kona u ya fhethu ha nnḓa tshifhinga tshi si ho fhasi ha awara dza 8 nga ḓuvha hu masiari.

Hune ha shumiswa mabogisi a zwiṱaha zwa khuhu nga tshithihi, hu vhe na tshiṱaha tshithihi kana u fhira kha khuhu dza tsadzi dza 8.

Khuhu dza tsadzi dzi ṱoḓa tshifhinga tshi si ho fhasi ha awara dza 8 dza swiswi ḽi sa ṱhukhuwi kha mutevhe muṅwe na muṅwe wa awara dza 24 uri dzi awele zwavhuḓi.

U dzudza fhasi nḓuni zwi ṱoḓa zwa u andadza kha tshipiḓa tshithihi kha zwiraru (33%) tsha vhupo ha fhasi kana u fhira.

Fhethu ha nnḓa ha u tshimbila a ho ngo tea u ḓala nga maanḓa: khuhu dzi songo fhira 5 nga mitara ya tshikwea ya fhethu afho.

An owner or manager whose animals are, or may be, infected with a controlled animal disease must report it to the state veterinary director immediately.

Newcastle disease and notifiable avian influenza are on the state list of controlled and notifiable animal diseases.

No animal may be slaughtered anywhere but an abattoir, with the exceptions below.

A i athu ṱolwa nga muḓivhi wa zwa vhulimi — ndi tsumbanḓila ya u pulana, a si ndaela. I sumbedzeni muṱuṱuwedzi waṋu wa zwa vhulimi, ni ite ndingo ya mavu musi ni sa athu shumisa tshelede kha mavu.
Ṱhalutshedzelo ya mvetamveto: a i athu u sedzwa nga muthu a ambaho luambo zwavhuḓi. Sedzani tshiko tsha Luisimane phanḓa ha u tevhedza zwine zwa ambiwa.

SYNTHETIC GRAPH SOURCE {COUNT} $&

Zwine zwa nga nangwa zwa bulasi - zwi khou bvela phanḓa

Shumisani zwe na zwi vhona bulasini na mitevhe mipfufhi ya ṱhoḓisiso. Khwaṱhisedzani zwine na nanga henefho musi ni sa athu u renga.

Zwibveledzwa zwa zwifuwo na ṱhogomelo (zwi khou bvela phanḓa)

Khuhu: Laying hens (commercial) (zwi khou bvela phanḓa)

Slaughter for your own household, or for cultural or religious purposes, is exempt from the abattoir rule.

Tshiko: SAPA Abridged Code of Practice: Commercial Layers, p. 1

Tshiko: Animal Diseases Act 35 of 1984, s.11(1)(b)(ii), p. 9

Tshiko: Department of Agriculture, Forestry and Fisheries, Controlled and Notifiable Animal Diseases (factsheet), p. 1

Tshiko: Meat Safety Act 40 of 2000, s.7(1)(a), p. 6

Tshiko: Meat Safety Act 40 of 2000, s.7(2)(a), p. 6

Tshiko: SAPA Industry Profile 2024, Table 4: commercial standards cumulative to 90 weeks, p. 39

Tshiko: SAPA Layer Notes, Commercial Layers (training module, November 2013), p. 13

Tshiko: SAPA Industry Profile 2022, Table 4a (Breed standards in laying hens), p. 39

Tshiko: ARC Climate-Smart Agriculture Training Manual, Poultry production, p. 8

Tshiko: ARC Climate-Smart Agriculture, module 11, printed pp. 468–480

Tshiko: FAO: Small-scale poultry production, egg marketing

Tshiko: SAPA Layer Notes, Commercial Layers (training module, November 2013)

A i athu ṱolwa nga muḓivhi wa zwa vhulimi — ndi tsumbanḓila ya u pulana, a si ndaela. I sumbedzeni muṱuṱuwedzi waṋu wa zwa vhulimi, ni ite ndingo ya mavu musi ni sa athu shumisa tshelede kha mavu.
Ṱhalutshedzelo ya mvetamveto: a i athu u sedzwa nga muthu a ambaho luambo zwavhuḓi. Sedzani tshiko tsha Luisimane phanḓa ha u tevhedza zwine zwa ambiwa.

SYNTHETIC GRAPH SOURCE {COUNT} $&

Zwine zwa nga nangwa zwa bulasi - zwi khou bvela phanḓa

Shumisani zwe na zwi vhona bulasini na mitevhe mipfufhi ya ṱhoḓisiso. Khwaṱhisedzani zwine na nanga henefho musi ni sa athu u renga.

Zwibveledzwa zwa zwifuwo na ṱhogomelo (zwi khou bvela phanḓa)

Masekwa: Meat duck (Pekin / Muscovy, backyard)

Zwifhaṱo zwa 1 zwi re hone; zwa 2 zwo dzinginywaho. Bindu ḽo nangwaho: ṉama. A hu humbulelwi tshivhalo tsha zwifuwo kana mvelelo.

Nyeletshedzo ya u renga: yelanyisani lushaka/zwifuwo na tshibveledzwa, furu ya henefho, kilima na vhukoni ha ṱhogomelo; khwaṱhisedzani na mueletshedzi wa henefho wa zwifuwo kana u fuwa ṉotshi.

Ndaedzi ya tshibveledzwa tsha u thoma: vhege dza 7–12. P44 = Pekin (7 weeks to slaughter); MR71 = Muscovy (10 weeks for ducks/hens, 12 weeks for drakes). Treated as 'time to first product' = age to slaughter, since this is a single-cycle meat enterprise (no repeat product like eggs). Non-South-African trial by Warsaw researchers, published in SAJAS; journal location does not establish local applicability.

Ndaedzi ya vhubveledzi: 2.3–2.9 kg ya tshileme tsha tshiṉoni tshi tshilaho/nga tshiṉoni musi tshi tshi ṱhavhiwa. Non-SA (Guyana study reporting a citation to Steklenev 1990); no SA-specific absolute slaughter live-weight figure was found (see gaps). 2.93 kg = Pekin at 8-10 weeks, 2.29 kg = Muscovy at 8-10 weeks - matches this enterprise's two breeds. The same LRRD paper's own experiment (Table 2, intensive system, Guyana) measured live weight at slaughter (8 weeks) as: Pekin drakes 2.53 kg, Pekin hens 2.29 kg, Muscovy drakes 2.47 kg, Muscovy hens 1.51 kg - broadly consistent with, but not merged into, the cited range above since it is a separate dataset/table in the same source. No sourced eggs/duck/year figure was found for either breed (see gaps); the LRRD and SAJAS sources used here studied ducks reared only for meat.

Ṱolani ndaedzi dzo ṱanḓavhuwaho dza bindu dza zwiḽiwa na madzulo. Yelanyisani mukovhe wa zwiḽiwa na tshikhala na tshiimo tsha tshifuwo na sisṱeme ya madzulo; ṉeani maḓi o kunaho a fulufhedzeaho.

Ndi mulandu u dzima sekwa zwiḽiwa kana maḓi, kana u ḽi vhea hune ha shaya tshikhala tsho eḓanaho, muya wo eḓanaho, tshedza kana vhudzumbamo kha mufhiso na phepho.

Sedzani miṱambi u vhona u fa nga khathihi, mithenga yo ṱhilinganyaho, u zwimba ha zwiṉama zwo nembelelaho fhasi ha mulomo kana kha ṱhoho, kana u fhungudzea ha makumba nga khathihi - hezwi ndi tsumbo dza mugunyuso wa zwiṉoni nahone dzi tea u vhigiwa nga u ṱavhanya.

Veani masekwa kule na zwiṉoni zwa ḓaka na maḓi kana zwiḽiwa zwo vuleaho zwi zwi kokodzaho, ni lange u dzhena ha vhaeni nḓuni dza masekwa, u fhungudza khombo ya mugunyuso wa zwiṉoni.

Ducks are one of the domesticated animal species listed in the Meat Safety Act, so its rules on registered abattoirs and hygiene apply to them.

Ducks slaughtered to be sold must be slaughtered at a registered abattoir; slaughter for the farmer's own consumption is exempt but that meat may not be sold.

At the abattoir, ducks must be handled humanely during loading, transport, offloading, housing, immobilising and killing, under the Animals Protection Act.

A i athu ṱolwa nga muḓivhi wa zwa vhulimi — ndi tsumbanḓila ya u pulana, a si ndaela. I sumbedzeni muṱuṱuwedzi waṉu wa zwa vhulimi, ni ite ndingo ya mavu musi ni sa athu shumisa tshelede kha mavu.
Ṱhalutshedzelo ya mvetamveto: a i athu u sedzwa nga muthu a ambaho luambo zwavhuḓi. Sedzani tshiko tsha Luisimane phanḓa ha u tevhedza zwine zwa ambiwa.

SYNTHETIC GRAPH SOURCE {COUNT} $&

Zwine zwa nga nangwa zwa bulasi - zwi khou bvela phanḓa

Shumisani zwe na zwi vhona bulasini na mitevhe mipfufhi ya ṱhoḓisiso. Khwaṱhisedzani zwine na nanga henefho musi ni sa athu u renga.

Zwibveledzwa zwa zwifuwo na ṱhogomelo (zwi khou bvela phanḓa)

Masekwa: Meat duck (Pekin / Muscovy, backyard) (zwi khou bvela phanḓa)

Ducks are birds/poultry, so an outbreak of avian influenza in a duck flock is a controlled animal disease under the Animal Diseases Act and must be reported to the state vet immediately.

Tshiko: Animals Protection Act 71 of 1962 (consolidated text, s.2(1)(b)-(c)), p. 2

Tshiko: Western Cape Department of Agriculture, Avian Influenza (Bird Flu) factsheet, p. 2

Tshiko: Meat Safety Act 40 of 2000, Schedule 1 (Animals to which this Act applies), p. 16

Tshiko: Meat Safety Act 40 of 2000, s.7, p. 6

Tshiko: Meat Safety Act 40 of 2000, s.9, p. 7

Tshiko: Damaziak et al. (2014), Influence of housing system on the growth and histological structure of duck muscles, S. Afr. J. Anim. Sci. 44(2): Poland trial

Tshiko: Livestock Research for Rural Development 18(11) 2006 - Solomon et al. (Guyana; not an SA figure), citing Steklenev 1990

A i athu ṱolwa nga muḓivhi wa zwa vhulimi — ndi tsumbanḓila ya u pulana, a si ndaela. I sumbedzeni muṱuṱuwedzi waṋu wa zwa vhulimi, ni ite ndingo ya mavu musi ni sa athu shumisa tshelede kha mavu.
Ṱhalutshedzelo ya mvetamveto: a i athu u sedzwa nga muthu a ambaho luambo zwavhuḓi. Sedzani tshiko tsha Luisimane phanḓa ha u tevhedza zwine zwa ambiwa.

SYNTHETIC GRAPH SOURCE {COUNT} $&

Zwine zwa nga nangwa zwa bulasi - zwi khou bvela phanḓa

Shumisani zwe na zwi vhona bulasini na mitevhe mipfufhi ya ṱhoḓisiso. Khwaṱhisedzani zwine na nanga henefho musi ni sa athu u renga.

Zwibveledzwa zwa zwifuwo na ṱhogomelo (zwi khou bvela phanḓa)

Tshivhudu: Tilapia (Mozambique tilapia, small pond or tank)

Zwifhaṱo zwa 1 zwi re hone; zwa 2 zwo dzinginywaho. Bindu ḽo nangwaho: khovhe. A hu humbulelwi tshivhalo tsha zwifuwo kana mvelelo.

Nyeletshedzo ya u renga: yelanyisani lushaka/zwifuwo na tshibveledzwa, furu ya henefho, kilima na vhukoni ha ṱhogomelo; khwaṱhisedzani na mueletshedzi wa henefho wa zwifuwo kana u fuwa ṋotshi.

Ndaedzi ya tshibveledzwa tsha u thoma: vhege dza 43.5–52.2. 10 months x 4.35 = 43.5 weeks; 12 months x 4.35 = 52.2 weeks. This is Mozambique Tilapia (O. mossambicus) grow-out time to harvestable size; the source separately states Nile Tilapia (O. niloticus) reaches harvestable maturity faster, in 6-10 months (= 26.1-43.5 weeks) — not used here as it is a different species with a different legal status (see 'legal').

Ndaedzi ya vhubveledzi: 800 g ya tshileme musi i tshi kaṋiwa/nga khovhe. Table 8-3 gives the Mozambique Tilapia's harvestable size as an approximate single figure ('+- 800g'), not a range, so [800, 800] is used rather than inventing a spread. The same table cell wraps '10-12' across a line break in the PDF ('10-\n12'), kept verbatim. The table's Nile Tilapia row gives the same approximate weight (+-800g) reached faster, in 6-10 months — not used here because this dossier is framed around the indigenous, legally unrestricted Mozambique Tilapia per task instruction; see 'legal' for why Nile Tilapia is treated separately. No SA source was found giving a feed-conversion-based or per-hectare yield figure specific to O. mossambicus that could be converted to this per-fish unit without stitching numbers across documents, so outputUnit follows the source's own unit (grams harvest weight per fish) rather than being converted.

Ndaedzi ya vhukale/sisṱeme: ARC makes tilapia suitability depend on water temperature: summer grow-out can be suitable anywhere in South Africa when water exceeds 20°C; growth is poor below 18°C and winter water below 13°C can kill farmed tilapia. The Highveld reference is a summer crop; tunnels can widen the range. Measure pond water and confirm stocking species and dates. Air climate or a mapped pond does not prove harvest months, fish numbers or food kg.

Ṱolani ndaedzi dzo ṱanḓavhuwaho dza bindu dza zwiḽiwa na madzulo. Yelanyisani mukovhe wa zwiḽiwa na tshikhala na tshiimo tsha tshifuwo na sisṱeme ya madzulo; ṋeani maḓi o kunaho a fulufhedzeaho.

The indigenous Mozambique Tilapia (O. mossambicus) is not a listed invasive species and may be farmed without an alien-species permit; the faster-growing Nile Tilapia (O. niloticus) is.

Nile Tilapia (Oreochromis niloticus) is a nationally listed invasive fish species: Category 1b inside National Parks, Provincial Reserves, Mountain Catchment Areas and Forestry Reserves, and Category 3 in all other areas.

Outside protected areas Nile tilapia is Category 3: it may remain in prescribed areas or provinces, but further propagation or trade is prohibited.

Nile Tilapia has been listed under the NEMBA Alien and Invasive Species Regulations as requiring a permit to produce.

Producing an alien or invasive species (such as Nile Tilapia) is prohibited in areas designated under the Protected Areas Act and the Fish Sanctuary Act; a permit is required to produce it elsewhere.

A i athu ṱolwa nga muḓivhi wa zwa vhulimi — ndi tsumbanḓila ya u pulana, a si ndaela. I sumbedzeni muṱuṱuwedzi waṋu wa zwa vhulimi, ni ite ndingo ya mavu musi ni sa athu shumisa tshelede kha mavu.
Ṱhalutshedzelo ya mvetamveto: a i athu u sedzwa nga muthu a ambaho luambo zwavhuḓi. Sedzani tshiko tsha Luisimane phanḓa ha u tevhedza zwine zwa ambiwa.

SYNTHETIC GRAPH SOURCE {COUNT} $&

Zwine zwa nga nangwa zwa bulasi - zwi khou bvela phanḓa

Shumisani zwe na zwi vhona bulasini na mitevhe mipfufhi ya ṱhoḓisiso. Khwaṱhisedzani zwine na nanga henefho musi ni sa athu u renga.

Zwibveledzwa zwa zwifuwo na ṱhogomelo (zwi khou bvela phanḓa)

Tshivhudu: Tilapia (Mozambique tilapia, small pond or tank) (zwi khou bvela phanḓa)

Producing tilapia in South Africa requires a water use licence under the National Water Act.

Tshiko: IDC, Research into the Potential for the Production, Processing and Export of Tilapia for the Southern African Market (2015), Table 8-3 (Mozambique vs. Nile Tilapia), p. 69

Tshiko: Dept. of Environmental Affairs / invasives.org.za, National Listed Invasive Species (NEMBA AIS Regulations booklet), List 7 (National List of Invasive Fresh-water Fish Species), p. 32

Tshiko: Dept. of Environmental Affairs / invasives.org.za, National Listed Invasive Species (NEMBA AIS Regulations booklet), p. 3

Tshiko: IDC, Research into the Potential for the Production, Processing and Export of Tilapia for the Southern African Market (2015), p. 69

Tshiko: IDC, Research into the Potential for the Production, Processing and Export of Tilapia for the Southern African Market (2015), p. 74

Tshiko: IDC, Research into the Potential for the Production, Processing and Export of Tilapia for the Southern African Market (2015), p. 75

Tshiko: ARC Aquaculture Production App: General FAQ, Can I grow tilapia on the Highveld?

Tshaka dza zwiliṅwa dza u sedzulusa henefho

Vhulungani dzina ḽa lushaka lune na nanga zwa vhukuma. Mitevhe mipfufhi ya ṱhoḓisiso i ṱoḓa khwaṱhisedzo ya khalaṅwaha ya henefho, maḓi na mavu.

Tomatoes

Lushaka lwo rekhodwaho: a lu athu u nangwa.

Kilima ya fhethu hafha a i athu u ṱalutshedzea, ngauralo a hu buliwi mutevhe mupfufhi wa henefho. Vulani garaṱa ya tshiliṅwa u vhona zwiko na u rekhoda zwe na nanga.

A i athu ṱolwa nga muḓivhi wa zwa vhulimi — ndi tsumbanḓila ya u pulana, a si ndaela. I sumbedzeni muṱuṱuwedzi waṋu wa zwa vhulimi, ni ite ndingo ya mavu musi ni sa athu shumisa tshelede kha mavu.
Ṱhalutshedzelo ya mvetamveto: a i athu u sedzwa nga muthu a ambaho luambo zwavhuḓi. Sedzani tshiko tsha Luisimane phanḓa ha u tevhedza zwine zwa ambiwa.